

Discrete Probability Distribution

A discrete random variable assumes each of its values with a certain probability. It is convenient to represent all the probabilities of a random variable, X by a formula.

$$\boxed{f(x) = P(X = x)}$$

The set of ordered pairs $(x, f(x))$ is a probability function, probability mass funcⁿ (PMF) or probability distribuⁿ of the discrete random variable X , if the following rules sustain for each possible outcome x .

$$1. f(x) \geq 0$$

$$2. \sum_x f(x) = 1$$

$$3. P(X = x) = f(x)$$

There are many problems where we may wish to compute the probability that the observed value of a random variable X will be less than or equal to some real no. x .

The CDF or cumulative distribution funcⁿ $F(x)$ of a discrete random variable X with

PMF $f(x)$ is

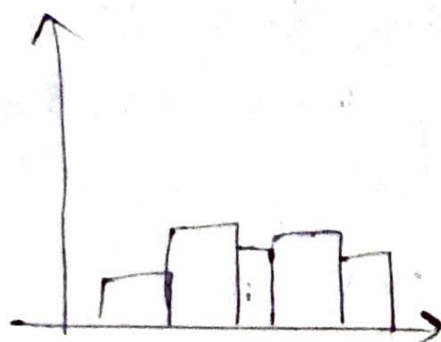
$$F(x) = P(X \leq x) = \sum_{t \leq x} f(t)$$

for $-\infty < x < \infty$

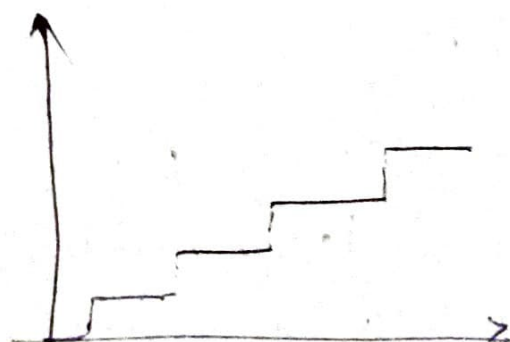
- We can also choose to represent a probability distribution in a graphical format. Sometimes, as probability histograms $\rightarrow$ the bases are constructed in a way that leaves no space between the rectangles.



PMF Plot



Probability histogram



discrete CDF